ZIFENG (STEVEN) ZHOU

(520) 339-3450 | zifengzhou01@gmail.com | zifeng-zhou.github.io

EDUCATION

Rice University — **Houston, TX**
Master of Computer Science — *Aug 2026 - Dec 2027*
University of Arizona — ***Tucson, AZ***
B.S. in Computer Science, cum laude — *Sep 2023 - May 2026*
• **Honors:** Dean's List (2x); Dean's List with Distinction

SKILLS

Languages: Python, C++, C, JavaScript
Machine Learning: PyTorch, Hugging Face Transformers, XGBoost, OpenCV, Stable Diffusion, LoRA
AI/ML Areas: LLM Evaluation, Adversarial Robustness, Generative AI, Computer Vision, RLHF
Development & Data: Pandas, NumPy, Git, Linux, SSH, Jupyter, Flask, REST APIs

RESEARCH EXPERIENCE

Institute of Automation, Chinese Academy of Sciences (CASIA) — Remote
Research Assistant, LLM Safety & RLHF — *Nov 2025 - Present*
• **LLM alignment vs. robustness:** Studied whether safety-aligned LLMs remain reliable under adversarial attacks; built a reproducible evaluation pipeline to compare multiple open-source models across benign, prefix-injection, and adaptive attack settings.
• **Robustness metrics and findings:** Designed metrics including Prefix-k ASR curves, Robustness AUC, and Adaptivity Gap; identified cases where stronger benign alignment did not translate into stronger adversarial robustness, including ranking reversals under out-of-distribution attacks.
• **RLHF analysis:** Analyzed reward modeling, RLHF, Q-value estimation, and KL regularization to inform experimental design and interpretation of alignment behavior.
• **Research manuscript:** Preparing a **first-author manuscript for ICLR 2027** on trustworthy LLM evaluation and adversarial robustness.

University of Arizona — **Tucson, AZ**
Research Assistant, Language Agent Tree Search — *Mar 2025 - Sep 2025*
• **LLM reasoning evaluation:** Built model-evaluation workflows with Python, PyTorch, and Hugging Face Transformers to evaluate LATS and baseline reasoning methods on AIME and HotPotQA.
• **Experimentation and debugging:** Improved experiment reliability through checkpoint logging, structured debugging, and automated result analysis across multiple reasoning benchmarks.
• **Results:** Observed 20%+ accuracy improvement on selected discrete mathematics and logical reasoning tasks, with stronger performance on HotPotQA multi-hop reasoning compared with baseline methods.

PROFESSIONAL EXPERIENCE

Microsoft — Remote
Artificial Intelligence Intern — *Nov 2025 - Jan 2026*
• **Machine learning:** Built and evaluated XGBoost, TabNet, and SVM models for enterprise valuation and loan-risk analysis, using iterative testing and validation to assess model performance and stability.
• **ML engineering:** Established reproducible Windows/Linux development environments using Git, SSH, Miniconda, GitHub Codespaces, and cloud GPU resources.
• **Generative AI:** Explored Stable Diffusion and LoRA workflows for text-to-image and text-to-video generation and documented reproducible experiment configurations on GitHub.

Global Impact Assessment (GIA) — Remote
Artificial Intelligence Intern — *Sep 2025 - Nov 2025*
• **Computer vision:** Processed post-disaster satellite imagery from the xBD dataset using Python, OpenCV, and PyTorch; contributed to training and evaluation of U-Net and RetinaNet models.
• **Change detection:** Implemented pre/post-disaster change detection using pixel difference and SSIM and analyzed failure cases caused by preprocessing and complex terrain.
• **Visualization and reporting:** Integrated CNN predictions with geographic data to estimate affected populations, built Folium-based interactive damage maps, generated LLM-assisted disaster summaries, and wrote scripts for automated HTML comparison reports.

PROJECTS & ADDITIONAL EXPERIENCE

AI Medical Consultation Application — **Tucson, AZ**
Co-Developer, University of Arizona — *Jun 2025 - Aug 2025*
• **Application development:** Built React and Material UI components for an AI medical consultation platform, including dashboard navigation, reporting controls, and role-based access management.
• **Backend integration:** Integrated RESTful APIs for secure login, session tracking, and user-specific data loading; used Postman and GitHub workflows for API testing, debugging, and collaboration.
• **Delivery:** Helped deliver a test version supporting role-based module access and viewing/exporting virtual-patient chat records.

Undergraduate Teaching Assistant — **Tucson, AZ**
University of Arizona — *Aug 2025 - Dec 2025*
• **Teaching support:** Supported **100+ undergraduate students** through lab and review sessions, tutoring, regular Q&A, and explanations of Python programming concepts and problem-solving approaches.
• **Course operations:** Graded programming assignments and exams and assisted with exam proctoring under faculty supervision while maintaining academic integrity.